

Sentiment Analysis of Movie Reviews

End-to-End Data Science Pipeline & Web Deployment

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1. Executive Summary

This report details the successful development and deployment of a Machine Learning model designed to classify movie reviews into "Positive" or "Negative" sentiments. The project covers the entire data science lifecycle: from raw data extraction and natural language processing (NLP) to model training and final deployment via a Streamlit web application. The primary goal was to create a lightweight, highly accurate, and interactive sentiment analyzer.

2. Dataset Overview

The model was trained using the benchmark **IMDB Movie Reviews Dataset**, a widely recognized corpus for binary sentiment classification.

- **Total Records:** 50,000 distinct reviews.
- **Data Distribution:** Perfectly balanced (25,000 positive and 25,000 negative reviews).
- **Variables:** The dataset comprises two columns: `review` (the raw text) and `sentiment` (the target label).

3. Data Preprocessing & NLP Pipeline

Raw text data is unstructured and noisy, requiring rigorous cleaning before it can be fed into an algorithm. The following pipeline was implemented:

- **HTML Stripping:** Removed lingering web artifacts using regular expressions.
- **Noise Reduction:** Filtered out numbers, punctuation, and special characters to retain only alphabetic words.
- **Normalization:** Converted all text to lowercase to ensure uniformity (e.g., treating "Movie" and "movie" identically).
- **Stopwords Removal:** Utilized the Natural Language Toolkit (NLTK) library to eliminate common, non-informative English words (e.g., "is", "the", "and").

4. Feature Engineering & Modeling

Machine learning models require numerical input. To achieve this, the cleaned text was transformed using **TF-IDF (Term Frequency-Inverse Document Frequency)** Vectorization. This approach not only counts word frequencies but penalizes words that appear too frequently across all reviews, highlighting genuinely significant terms. The vocabulary was capped at the top 5,000 features to optimize processing speed and mitigate overfitting.

The dataset was split into an 80% training set (40,000 reviews) and a 20% testing set (10,000 reviews). **Logistic Regression** was selected as the classification algorithm due to its excellent baseline performance on sparse text data, interpretability, and execution speed.

5. Evaluation & Results

Final Model Accuracy: 88.92%

The Logistic Regression model demonstrated robust predictive capabilities on the unseen test data. The near 89% accuracy indicates that the model successfully learned the semantic patterns distinguishing positive praise from negative criticism. Precision and recall scores were balanced across both classes, ruling out any bias towards a specific sentiment.

6. Deployment & Deliverables

To transition the project from a static notebook to a functional product, the trained AI model (`sentiment_model.pkl`) and the TF-IDF vocabulary (`vectorizer.pkl`) were serialized using Python's Pickle library.

A web interface was subsequently developed using **Streamlit** (`app.py`). This interface allows end-users to input custom movie reviews into a text area and receive instant, real-time sentiment predictions. The environment configuration and dependency management were strictly handled in VS Code to ensure smooth local execution.

7. Conclusion

This project successfully demonstrates the practical application of NLP and predictive modeling. The transition from raw data to a fully functional, interview-ready web application fulfills the core requirements of the Navitech Data Science Capstone. The high accuracy and clean architecture ensure the project is scalable and maintainable for future enhancements.